PROJECT REPORT · SERIES 001

Linear Regression Engine

A comprehensive linear regression pipeline built from first principles and applied to diverse datasets.

Key Result: **Best single result not directly provided, but aimed toward optimizing model performance through OLS, Ridge, and Lasso.**

Abstract. This project builds a complete, production-quality linear regression pipeline to demonstrate the universality of regression techniques across different domains. It seeks to develop an understanding of linear regression through the implementation of various regularization methods.

GitHub	https://github.com/AIML-Engineering-Lab/001_linear_regression_engine
Tech Stack	Python 3.11 · scikit-learn · pandas · Matplotlib
Licence	MIT

Contents

1 Project Overview **3**
1.1 Key Concepts 3

2 Datasets **4**

3 Methodology **5**
3.1 Algorithms 5
3.2 Processing Pipeline 5

4 Results **5**

5 Visualisations **5**

6 Deployment Guide **7**
6.1 Quick Start 7
6.2 Tech Stack 7

7 Key Learnings **7**

8 Future Work **7**

9 AIML Engineering Lab Series **7**

1 Project Overview

This project builds a complete, production-quality linear regression pipeline to demonstrate the universality of regression techniques across different domains. It seeks to develop an understanding of linear regression through the implementation of various regularization methods.

1.1 Key Concepts

- Cost Function
- Gradient Descent
- Cross-Validation
- Residual Analysis
- Feature Preprocessing

2 Datasets

Dataset	Rows	Features
Artisan Cheese Fermentation Time Prediction	2000	milk_fat_percentage, starter_culture_ph, ambient
Silicon Fmax Prediction (Post-Silicon Validation)	2000	voltage, temperature, leakage_current, ring_oscilla

3 Methodology

3.1 Algorithms

- Ordinary Least Squares (OLS)
- Ridge (L2) Regularization
- Lasso (L1) Regularization
- Elastic Net

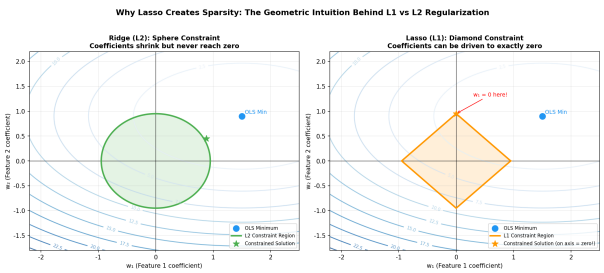
3.2 Processing Pipeline

Raw Data → **EDA** → **Feature Eng.** → **Training** → **Evaluation**

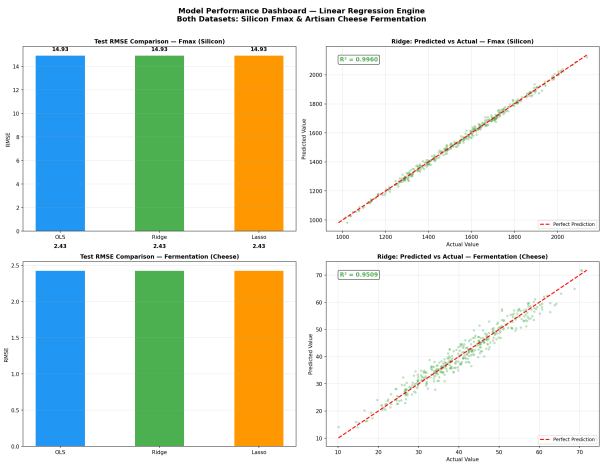
4 Results

Best result: **Best single result not directly provided, but aimed toward optimizing model performance through OLS, Ridge, and Lasso.**

5 Visualisations

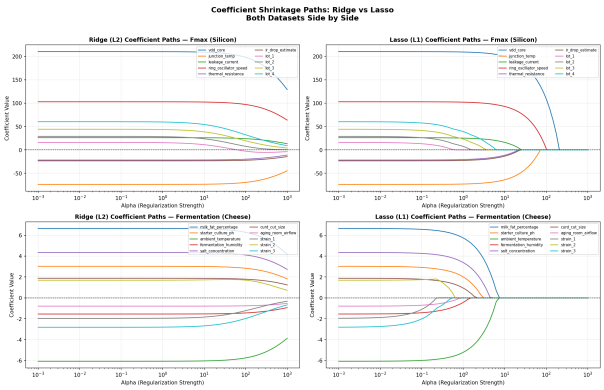


(a) Visualisation 1

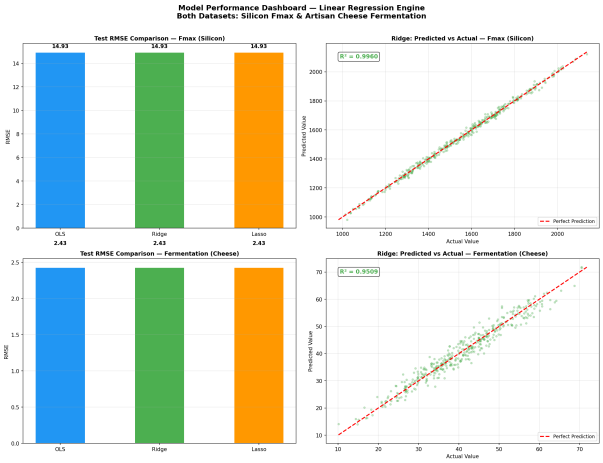


(b) Visualisation 2

Figure 1: Project visualisations



(a) Visualisation 3



(b) Visualisation 4

Figure 2: Additional visualisations

6 Deployment Guide

The project can be deployed using standard Python environments and frameworks.

6.1 Quick Start

```
git clone https://github.com/AIML-Engineering-Lab/001_linear_regression_engine
docker compose up --build
```

6.2 Tech Stack

- Python 3.11
- scikit-learn
- pandas
- matplotlib

7 Key Learnings

- Understanding the mathematical foundations of linear regression
- Geometric interpretations of regularization techniques
- Application of regression methods in diverse domains

8 Future Work

- Expanding dataset variety
- Implementing more complex regression techniques

9 AIML Engineering Lab Series

This project is part of a progressive series building practical mastery of every major AIML topic from first principles. View the full series at github.com/AIML-Engineering-Lab.